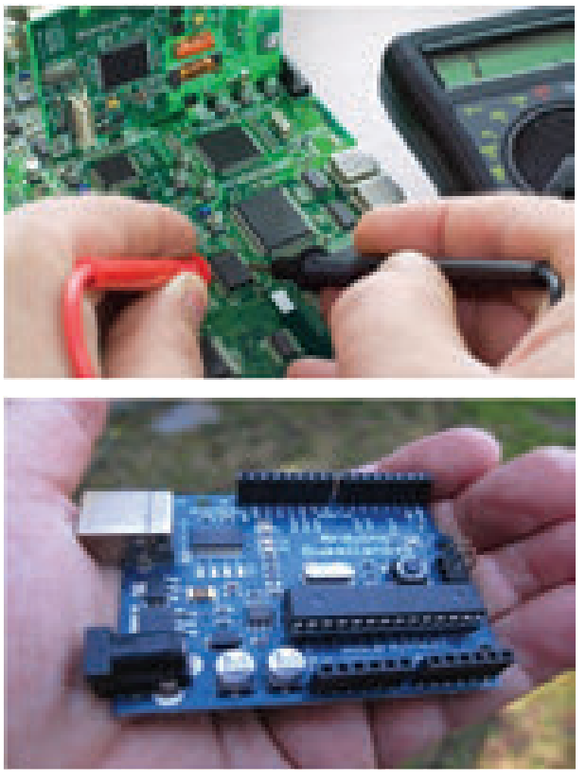


**Figure 10.49** Circuits with integrated chips

small size and low cost of ICs. ICs can function as an amplifier, oscillator, timer, microprocessor and computer memory.

These extremely small ICs can perform calculations and store data using either digital or analog technology. Digital ICs use logic gates, which work only with values of ones and zeros. A low signal sent to a component on a digital IC will result in a value of 0, while a high signal creates a value of 1.

**Digital ICs** usually find their applications in computers, networking equipment, and most consumer electronics. **Analog ICs** or linear ICs work with continuous values. This means a component on a linear IC can take any value and output another value. Linear ICs are typically used in audio and radio frequency amplification.

## 10.8 COMMUNICATION SYSTEMS

### Introduction

**Communication is the process of exchanging information by speaking, writing or using some other medium.**

Communication has existed since the dawn of life in this world. Growth in science and technology removed the locational disadvantage effectively. Information can be exchanged from one person to another anywhere on this Earth. Right from the researches done in communication by great scientists like J.C. Bose, G. Marconi and Alexander Graham Bell, communication has witnessed development by leaps and bounds. The communication industry is one of the largest in size and is the oldest since communication through telegraph (1844), telephone (1876), and Radio (1887) started centuries back.

The intensive research in the mid- and late nineteenth century has led to the development of long-distance transmission in the shortest possible time. However, the 20$^{th}$ century witnessed a leap over the development of communication, meeting the demands of speed and secured transfer of data. This section provides a glimpse of the basic concepts of electronic communication, some important communication systems and their applications.

## 10.9 MODULATION

The transmission of information through short distances does not require



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